LAB2 Report

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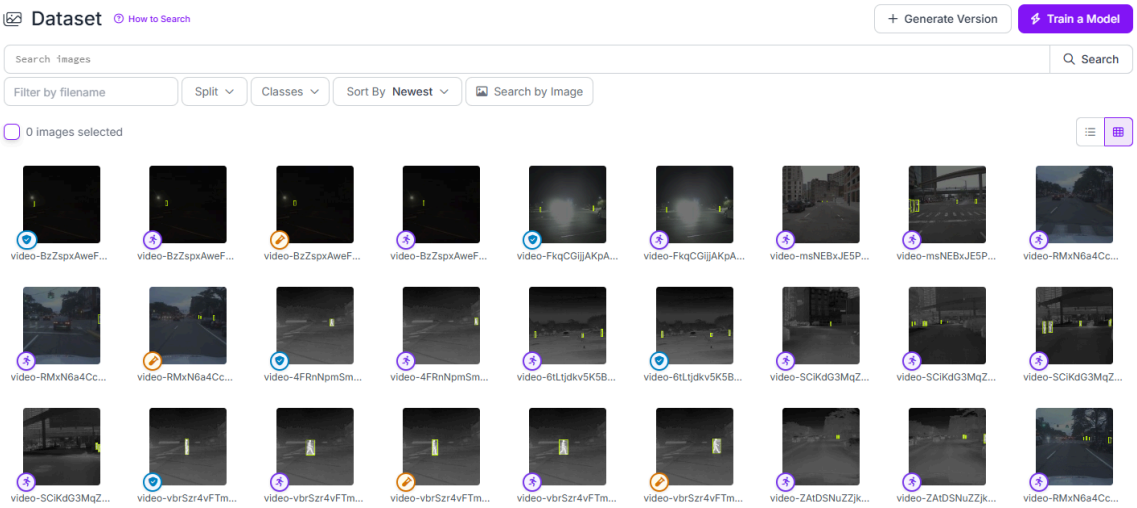
The application was made in order to detect pedestrians based on data from a video stream.

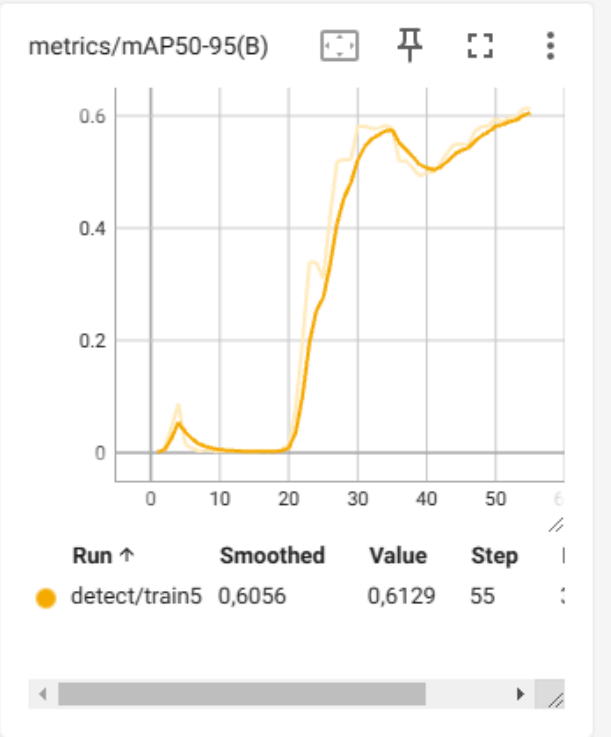
For this application, I used the yolov8n model, because it is by far the fastest to train, and can run very smoothly on colab.

To train the model, I first needed a dataset. I used Roboflow to annotate a small sample of images among the provided ones for the lab, and built a dataset off of them.



Then, I used the colab resources shared for the Lab and trained the yolov8 nano model on my dataset.

The metric used for evaluation is mAP. There are two different types of mAP measured by yolo: mAP50 and mAP50-95. After training, the mAP values of my model are as follow:



Here are the results from the validation set:

In conclusion, the model works well on the images given in the dataset, however after testing it on a custom video the results were quite poor. In order to increase the performance of the model, I think I would have to build a bigger dataset with very different images in order for the model to be able to detect pedestrians in any kind of environment.